

HARWELL RSX CHARACTERIZATION REPORT

OVERVIEW

Each resistor undergoes characterization at multiple temperature points to develop a precise model for calculating resistance across its operating range. Our modern resistors use a proprietary metal foil design that, depending on the unique assembly of each resistor, may be characterized via a polynomial or third-degree fit.

INSTRUMENT INFORMATION

Date: 13 NOV 2025
Manufacturer: Harwell
Model: RS-5
Serial Number: RS50052
Nominal Resistance: 1,000 Ω
Calibrated Resistance: 999.99080 Ω

TCR MODEL PARAMETERS

Calibrated Resistance: 999.99080 Ω (at 23°C)

Straight-Line TCR: -1.630 ppm/°C
Model Slope (at ref): -1.642 ppm/°C

Alpha (α): -1.642E-06 Ω/Ω/°C
Beta (β): -6.385E-09 Ω/Ω/°C²

RESISTANCE CALCULATION MODEL

To calculate the resistor value at a temperature other than the reference temperature (23.1°C), use this quadratic model:

$$R(T) = R_{(23)} + \alpha(T - 23) + \beta(T - 23)^2$$

Where:
T = Desired temperature (°C)
R₍₂₃₎ = 999.99080 Ω
α = -1.642E-03 Ω/°C
β = -6.385E-06 Ω/°C²

⚠ Note: This model may not be accurate outside the characterized temperature range.

RESISTANCE vs TEMPERATURE TABLE

Temperature (°C)	Temperature (°F)	Resistance (Ω)	Deviation (ppm)
18.0	64.4	999.998849	8.05
18.5	65.3	999.998059	7.26
19.0	66.2	999.997265	6.46
19.5	67.1	999.996468	5.67
20.0	68.0	999.995668	4.87
20.5	68.9	999.994864	4.06
21.0	69.8	999.994058	3.26
21.5	70.7	999.993248	2.45
22.0	71.6	999.992435	1.64
22.5	72.5	999.991619	0.82
23.0	73.4	999.990800	0.00
23.5	74.3	999.989978	-0.82
24.0	75.2	999.989152	-1.65
24.5	76.1	999.988323	-2.48
25.0	77.0	999.987491	-3.31
25.5	77.9	999.986656	-4.14
26.0	78.8	999.985817	-4.98
26.5	79.7	999.984976	-5.82
27.0	80.6	999.984131	-6.67
27.5	81.5	999.983283	-7.52
28.0	82.4	999.982432	-8.37

Temperature Coefficient of Resistance (TCR)

